

From working features to a memorable presentation.

16 FEATURE PRIORITIES

EXACT PAGES + REAL IMAGES

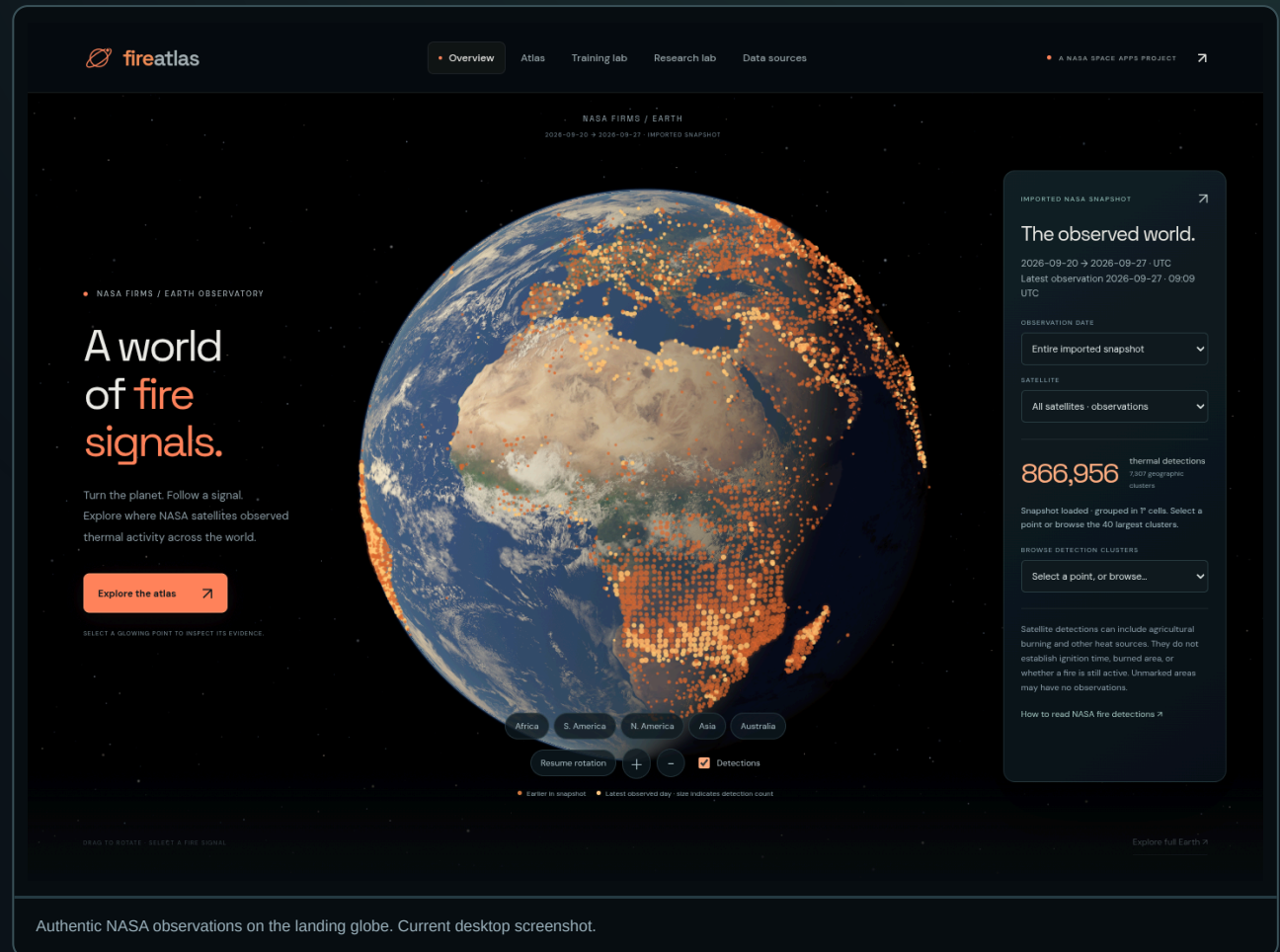
Polish one visible story at a time.

This guide inventories the project's strongest implemented features and gives each a concrete presentation target.

Use the page order as a work queue. Finish the interaction and its explanation before moving to the next feature.

"Standout" means valuable within this project. It does not claim that these capabilities are unique in the wider wildfire software market.

Screenshots show the current app. Improvements listed in this guide are recommendations, not completed redesigns.



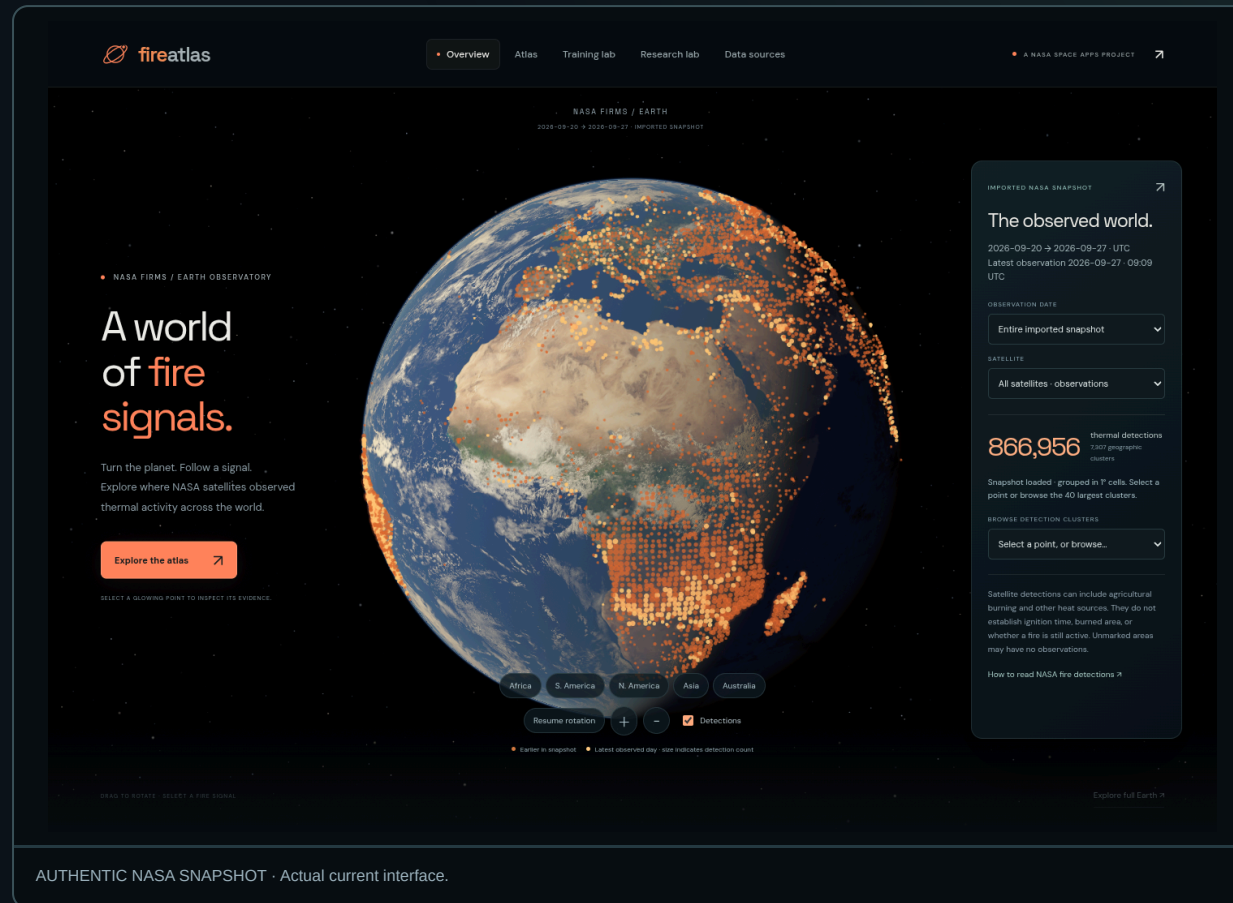
POLISH ROADMAP / FINISH EACH READY-TO-PRESENT CHECK

The order to work in.

ORDER	FEATURE	PDF PAGE	PRESENTATION ROLE
01	The interactive fire globe	03	First impression / core explanation
02	The burning activity calendar	04	First impression / core explanation
03	MODIS ↔ VIIRS comparison	05	First impression / core explanation
04	A daily observation replay	06	First impression / core explanation
05	Inspect the evidence behind a day	07	First impression / core explanation
06	Daily sensor overlap and ratio band	08	Depth / interactive proof
07	Candidate groups with adjustable assumptions	09	Depth / interactive proof
08	Observation exposure made visible	10	Depth / interactive proof
09	Commander incident rehearsal	11	Depth / interactive proof
10	The mobile Crew Device view	12	Depth / interactive proof
11	Disconnection and stale-data drills	13	Depth / interactive proof
12	Vegetation, cover and weather context	14	Supporting polish
13	Save, share and reproduce a study	15	Supporting polish
14	The five-stop guided story	16	Supporting polish
15	A separate window onto reported events	17	Supporting polish
16	Data provenance that a judge can inspect	18	Supporting polish

PRIORITY 01 / AUTHENTIC NASA SNAPSHOT

The interactive fire globe



Why it stands out

The visual entry point connects a planet-scale view directly to inspectable satellite evidence.

BEAUTIFY NEXT

- Give the selected point a clear halo and a compact place label.
- Keep the snapshot date readable beside the globe.
- Make the evidence card feel connected to the selected location; preserve mobile space for Earth.

READY TO PRESENT WHEN

A new visitor can select a signal, read its observation date and reach the matching atlas area in under 30 seconds.

866,956 imported detections in 7,307 display clusters. Counts are observations; ignition and current fire status are unknown.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/>

Interaction: Click a point → inspect its dates and satellites → open that area in the atlas.

PRIORITY 02 / SYNTHETIC SHOWCASE

The burning activity calendar



SYNTHETIC SHOWCASE · Actual current interface.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#calendar-section>

Interaction: Choose September 2026 → compare the monthly bars → select an active UTC day.

Why it stands out

The calendar translates point data into a readable season, with an explicit prior-year comparison.

BEAUTIFY NEXT

- Make the monthly peak, selected month and selected day visually distinct.
- Improve chart labels and hover/focus values before adding animation.
- Keep the three prior baseline years and unknown-coverage state near the main metric.

READY TO PRESENT WHEN

The audience can identify the seasonal peak and explain the baseline without opening documentation.

2026 examples have three comparable synthetic baseline years. Cell-days are a sampling measure, not a fire count.

PRIORITY 03 / SYNTHETIC SHOWCASE

MODIS ↔ VIIRS comparison



EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#calendar-section>

Interaction: Switch MODIS → VIIRS → Joint; watch the raw pixels and common cell-day values.

Why it stands out

This is the clearest demonstration of why different sensor counts need a shared comparison unit.

BEAUTIFY NEXT

- Emphasize the two input sensors and the shared-grid result with one visual sequence.
- Use consistent sensor colors across cards, calendar, map and research charts.
- Add a concise tooltip illustrating several pixels contributing to one cell-day.

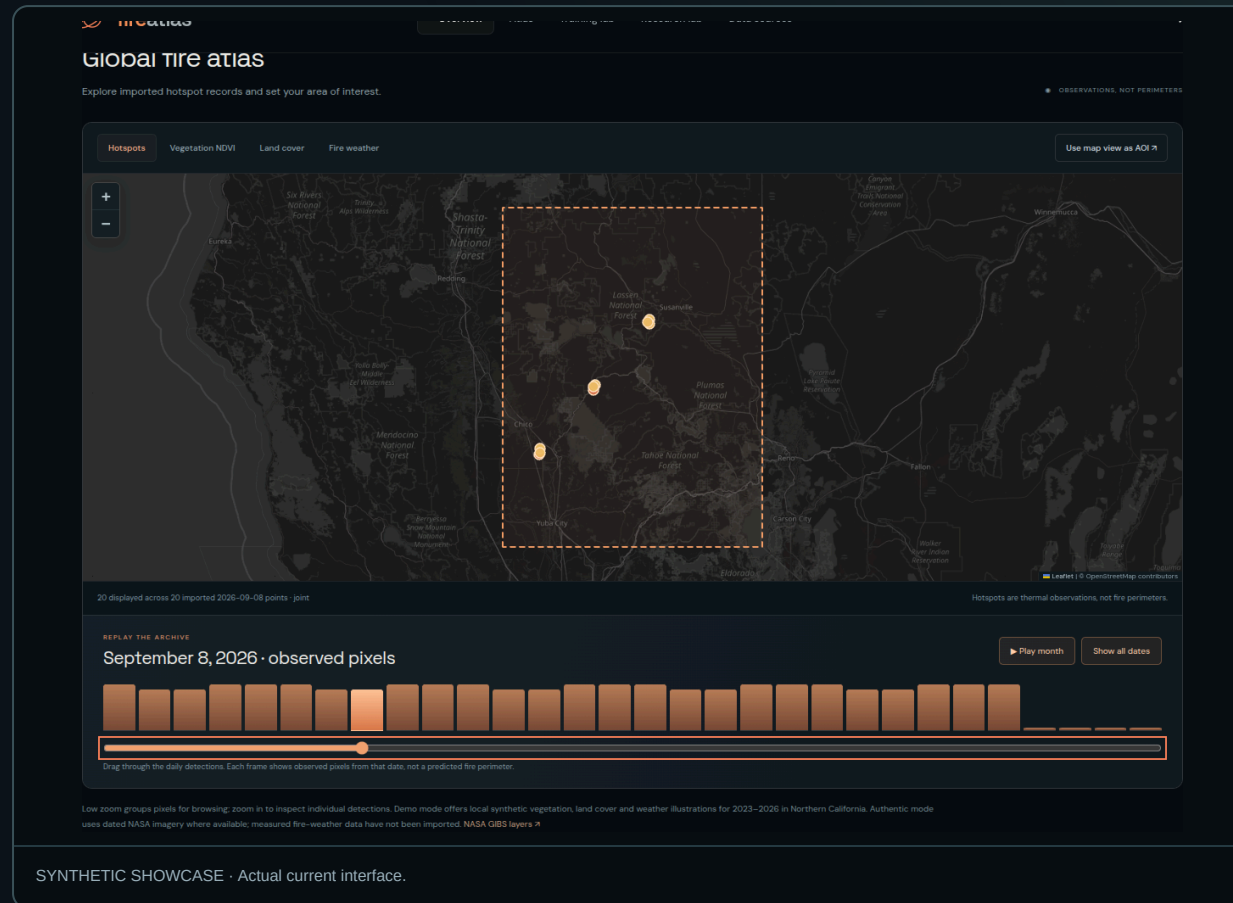
READY TO PRESENT WHEN

A viewer can explain why more VIIRS pixels do not automatically mean more distinct fires.

Raw pixels and joint cell-days have different units. This is not calibrated sensitivity or event deduplication.

PRIORITY 04 / SYNTHETIC SHOWCASE

A daily observation replay



EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#atlas-section>

Interaction: Drag the acquisition-day slider to day 8 → play the month → return to all dates.

Why it stands out

The map becomes a sequence the presenter can scrub through, rather than a static collection of dots.

BEAUTIFY NEXT

- Make the selected date large and keep playback controls together.
- Clarify the daily histogram and the difference between a day and all dates.
- Make map-to-calendar selection feel like one interaction with consistent highlighting.

READY TO PRESENT WHEN

Changing the day visibly changes the plotted observations and the audience can identify the active time filter.

This replays acquisition dates. It does not reconstruct the growth of a measured fire perimeter.

PRIORITY 05 / SYNTHETIC SCREEN / REAL PIPELINE

Inspect the evidence behind a day

05 / VERIFY

What is behind this day?

Inspect the original sensor records and their acquisition time.

2026-09-01 UTC

Source observations

22 RAW PIXELS

MODIS - T

12:30 UTC

confidence 62 · FRP 775

```
{
  "acq_date": "2026-09-01",
  "acq_time": "1230",
  "confidence": "62",
  "daynight": "D",
  "derivation": "Measured attributes reused; invented position/date/platform scenario. Not a NASA observation.",
  "frp": "7.75",
  "instrument": "MODIS",
  "latitude": "39.55000",
  "longitude": "-121.72000",
  "satellite": "T",
  "scan": "1.06",
  "seed_csv_line": "7",
  "seed_file_sha256": "bb27670859e2bb964514a2ea3118dea3c0ff64a4eecd76fe86558aaa01f760c8",
  "seed_filename": "MODIS_C6_1_Global_7d.csv",
  "synthetic": "true",
  "track": "1.03",
  "version": "presentation-v1"
}
```

READ THE SIGNAL CORRECTLY

A hotspot is an observation.

A cell-day counts a common grid cell with one or more detected thermal pixels on a UTC day. It is a sampling proxy. It does not count individual fires or burned area.

Why two sensors?

VIIRS can detect smaller hotspots than MODIS. Switch sources to compare their raw evidence; the joint view counts a shared cell/day once.

What does gray mean?

Missing source exports or unknown satellite pass/cloud coverage must not be read as a fire-free day.

SOURCE PROVENANCE

ease0933-centroid-lkm-v1

MODIS_SP · version presentation-v1

VIIRS_SNPP_SP · version presentation-v1

SYNTHETIC SCREEN / REAL PIPELINE · Actual current interface.

Why it stands out

Every headline number can lead to coordinates, observation time, sensor metadata and source provenance.

BEAUTIFY NEXT

- Replace dense record blocks with a clear time / location / sensor hierarchy.
- Keep raw confidence in the satellite's own scale and label FRP units.
- Make the selected day persistent while inspecting multiple records.

READY TO PRESENT WHEN

A presenter can trace one calendar value to at least one original source row without losing the selected date.

The pictured rows are labelled synthetic. Authentic imports use the same inspection workflow.

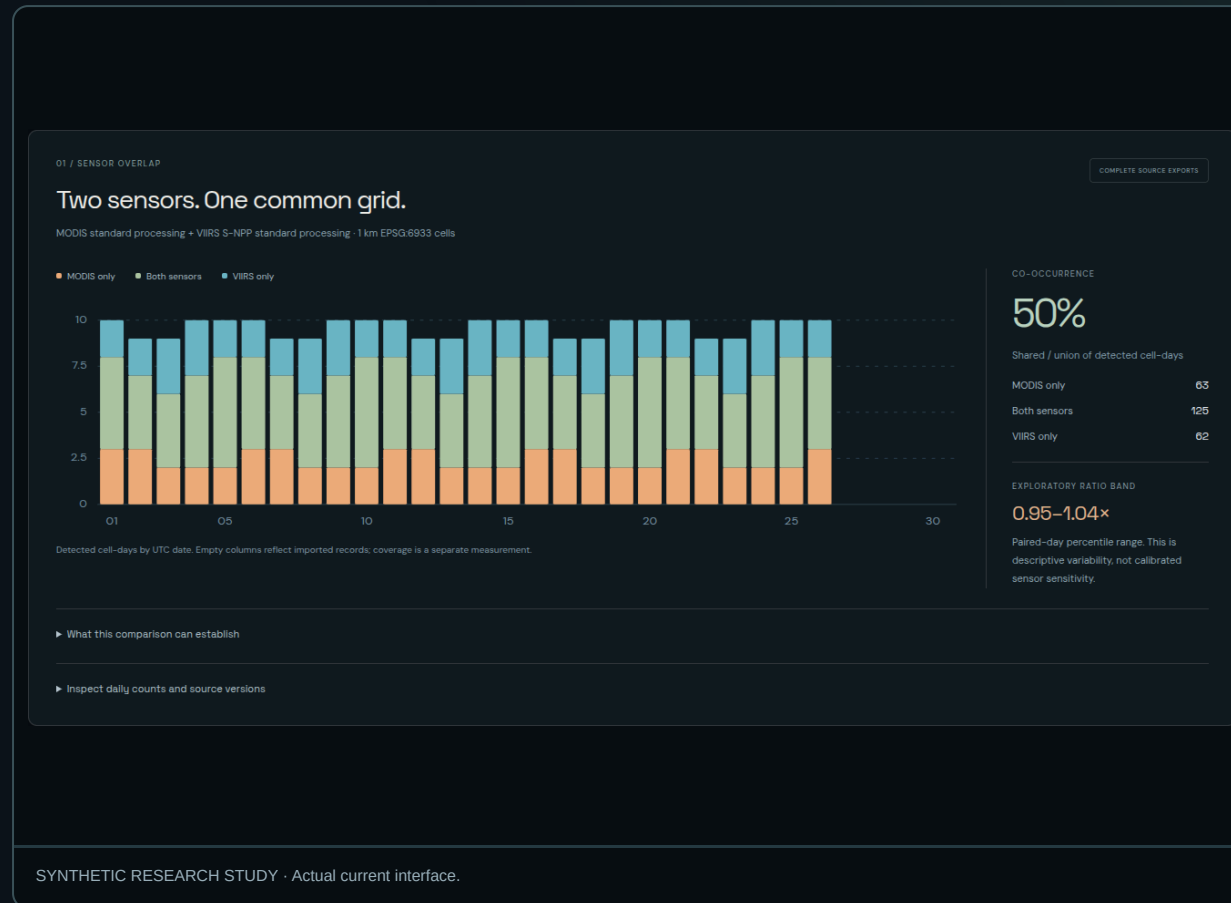
EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#evidence-section>

Interaction: Select an active calendar day first → expand the first source record.

PRIORITY 06 / SYNTHETIC RESEARCH STUDY

Daily sensor overlap and ratio band



Why it stands out

The project exposes what each sensor contributes and how much evidence is shared on the common grid.

BEAUTIFY NEXT

- Give the chart more space and strengthen the matching legend.
- Place the overlap percentage next to its numerator and denominator.
- Present the ratio band as a range with a short interpretation, keeping its exploratory label.

READY TO PRESENT WHEN

MODIS-only, shared and VIIRS-only observations are understandable at a glance.

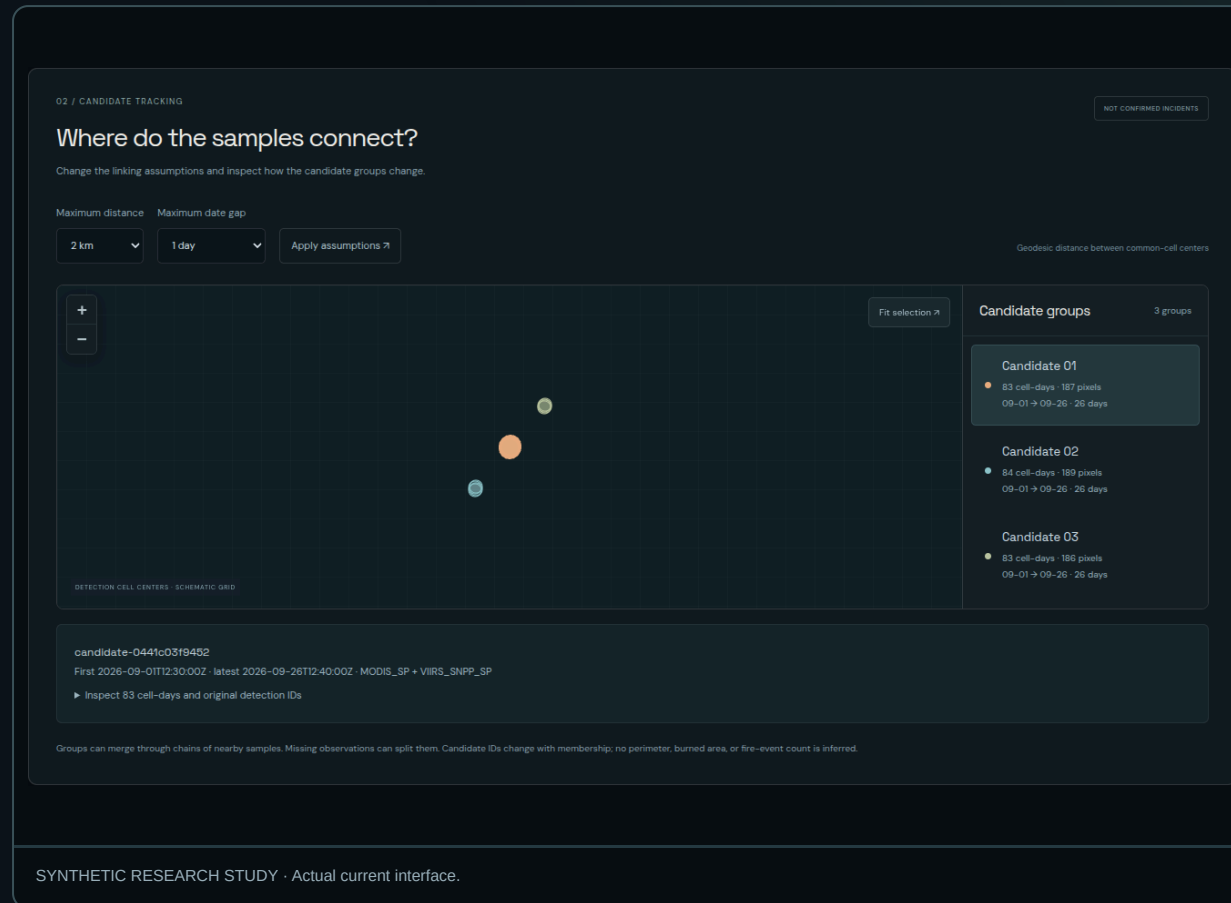
The default example has 125 shared cell-days and 50% overlap. The band is descriptive variability, not sensor calibration.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/research.html?demo=1&year=2026&month=9&exposure=synthetic>

Interaction: Open Research Lab → inspect the overlap chart and exploratory ratio band → export the study.

Candidate groups with adjustable assumptions



Why it stands out

Users can see how a grouping depends on its assumptions instead of accepting an unexplained incident count.

BEAUTIFY NEXT

- Make selected groups prominent on both map and list.
- Use a compact group summary with dates, cell-days and pixel count.
- Show a before/after group-count transition when assumptions change.

READY TO PRESENT WHEN

Changing the same-day/date-gap setting produces an explainable visual response and inspectable membership.

These are candidate detection groups. They are not confirmed wildfire incidents or measured boundaries.

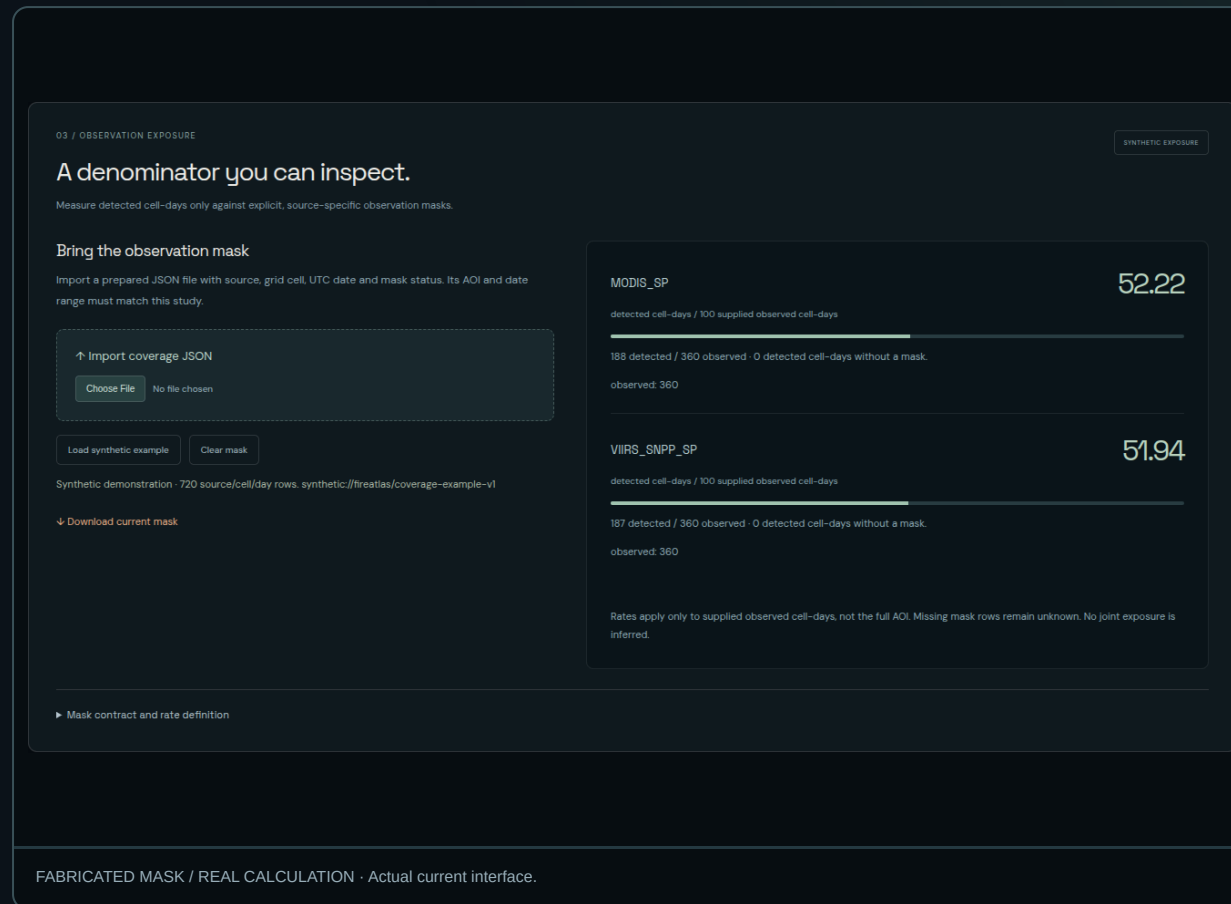
EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/research.html?demo=1&year=2026&month=9&exposure=synthetic>

Interaction: Scroll to Candidate tracking → change distance or date gap → Apply assumptions → inspect a group.

PRIORITY 08 / FABRICATED MASK / REAL CALCULATION

Observation exposure made visible



FABRICATED MASK / REAL CALCULATION · Actual current interface.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/research.html?demo=1&year=2026&month=9&exposure=synthetic>**Interaction:** Use the preloaded synthetic mask → compare sensor rates → clear the mask and show Unknown.

Why it stands out

The app distinguishes missing observation coverage from zero detected activity.

BEAUTIFY NEXT

- Visualize detected versus observed cell-days using a simple denominator diagram.
- Make the synthetic-mask state unmistakable without burying the result.
- Design the no-mask state as an informative explanation rather than an empty panel.

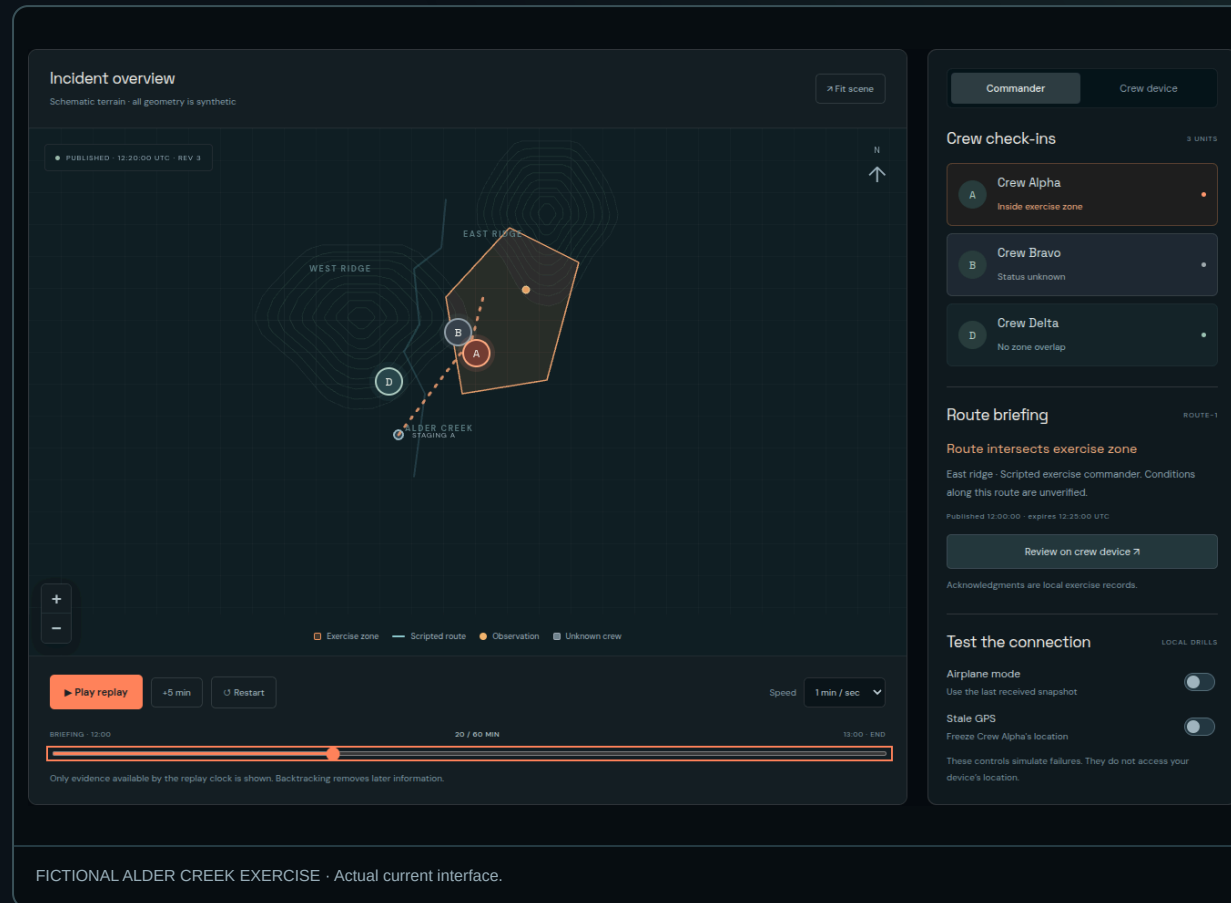
READY TO PRESENT WHEN

The presenter can explain why a missing coverage mask prevents a valid exposure rate.

The mask contains 720 fabricated source/cell/day rows. No real satellite cloud/pass mask has been imported.

PRIORITY 09 / FICTIONAL ALDER CREEK EXERCISE

Commander incident rehearsal



FICTIONAL ALDER CREEK EXERCISE · Actual current interface.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/training.html>

Interaction: Advance the replay to 12:20 UTC → inspect the zone, scripted route and crew check-ins.

Why it stands out

A second visible story connects evidence timing to a simulated team decision.

BEAUTIFY NEXT

- Prioritize replay time, received evidence and crew condition in the visual hierarchy.
- Use consistent map/list colors and legible route versus zone styling.
- Keep stale or disconnected crew visually distinct from fresh observations.

READY TO PRESENT WHEN

A viewer can identify what evidence is available now and which crew information is unknown.

All crew positions, zones and routes are fictional. Route approval and thresholds are scripted exercise inputs.

PRIORITY 10 / MOBILE BROWSER / SIMULATED CREW

The mobile Crew Device view



MOBILE BROWSER / SIMULATED CREW · Actual current interface.

EXACT PAGE / OPEN THIS VIEW<http://127.0.0.1:8000/training.html>**Interaction:** Tap Crew device → select Crew Bravo → inspect route briefing → acknowledge a briefing.

Why it stands out

A phone-sized crew screen lets the presenter demonstrate the same scenario from a different role.

BEAUTIFY NEXT

- Make current status the first screen element; enlarge primary touch targets.
- Put briefing, acknowledgment and connection state in a short action sequence.
- Reduce repeated prose and make uncertainty visible through labels as well as color.

READY TO PRESENT WHEN

On the actual presentation phone, the user can choose a crew, read the briefing and acknowledge it without horizontal scrolling.

This is a local simulated view. It does not synchronize with another phone or use the real device's GPS.

PRIORITY 11 / VERIFIED BROWSER OFFLINE RECOVERY

Disconnection and stale-data drills

PICK UP WHERE YOU LEFT OFF**Exercise restored · Progress saved in this tab · 0 local acknowledgments****Offline page ready · this tab can reload without a connection.****The replay pauses on reload. Progress stays in this tab; export your record before closing it. Local acknowledgments are not sent to another device.****Check offline readiness**

VERIFIED BROWSER OFFLINE RECOVERY · Actual current interface.

EXACT PAGE / OPEN THIS VIEW<http://127.0.0.1:8000/training.html>**Interaction:** Wait for Offline page ready → advance five minutes → disconnect → reload the same tab.

Why it stands out

The exercise explicitly demonstrates what happens when a connection or location update becomes unreliable.

BEAUTIFY NEXT

- Design one strong disconnected banner and a clear age indicator.
- Keep recovery, last received evidence and paused replay time together.
- Make reconnect and export actions easy to reach on a phone.

READY TO PRESENT WHEN

The prepared tab reloads offline at its saved clock; it clearly labels stale evidence and still exports the exercise record.

Offline recovery is scoped to a prepared Training Lab tab. A new tab cannot invent a saved briefing.

PRIORITY 12 / ILLUSTRATIVE LOCAL LAYERS

Vegetation, cover and weather context



EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#atlas-section>

Interaction: Select Vegetation NDVI → Land cover → Fire weather in demo mode.

Why it stands out

Layer switching shows how environmental context can complement observed thermal activity.

BEAUTIFY NEXT

- Give each layer a visible legend, date and unit/class explanation.
- Balance overlay opacity so hotspots and terrain remain readable.
- Use a persistent layer chip and preserve the selected source and month.

READY TO PRESENT WHEN

The presenter can identify the active layer and whether it is a synthetic illustration or measured imagery.

Local showcase layers are synthetic. The weather index uses arbitrary units, not computed FWI; authentic imagery depends on GIBS.

Save, share and reproduce a study

KEEP THE EVIDENCE

A view worth returning to.

Save your place, share a selection, or take the underlying study with you.

Copy view link ↗ Download study

NAME THIS VIEW

e.g. California - July comparison Save view Choose a saved view Open Remove

Saved on this browser only. Links use the receiving server's current data.

► Source ledger 24 source imports 12/12 complete months

FUNCTIONAL EXPORT WORKFLOW · Actual current interface.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/?demo=1&year=2026&month=9&series=joint&bbox=-122,39,-120,41#calendar-section>

Interaction: Name and save a view → copy its link → download a study ZIP → verify the bundle.

Why it stands out

A visual finding can be revisited and its counts independently reproduced from the exported inputs.

BEAUTIFY NEXT

- Group save, share and download by the user's intent.
- Show clear success feedback and a meaningful saved-view name.
- Summarize what is inside the study before the download.

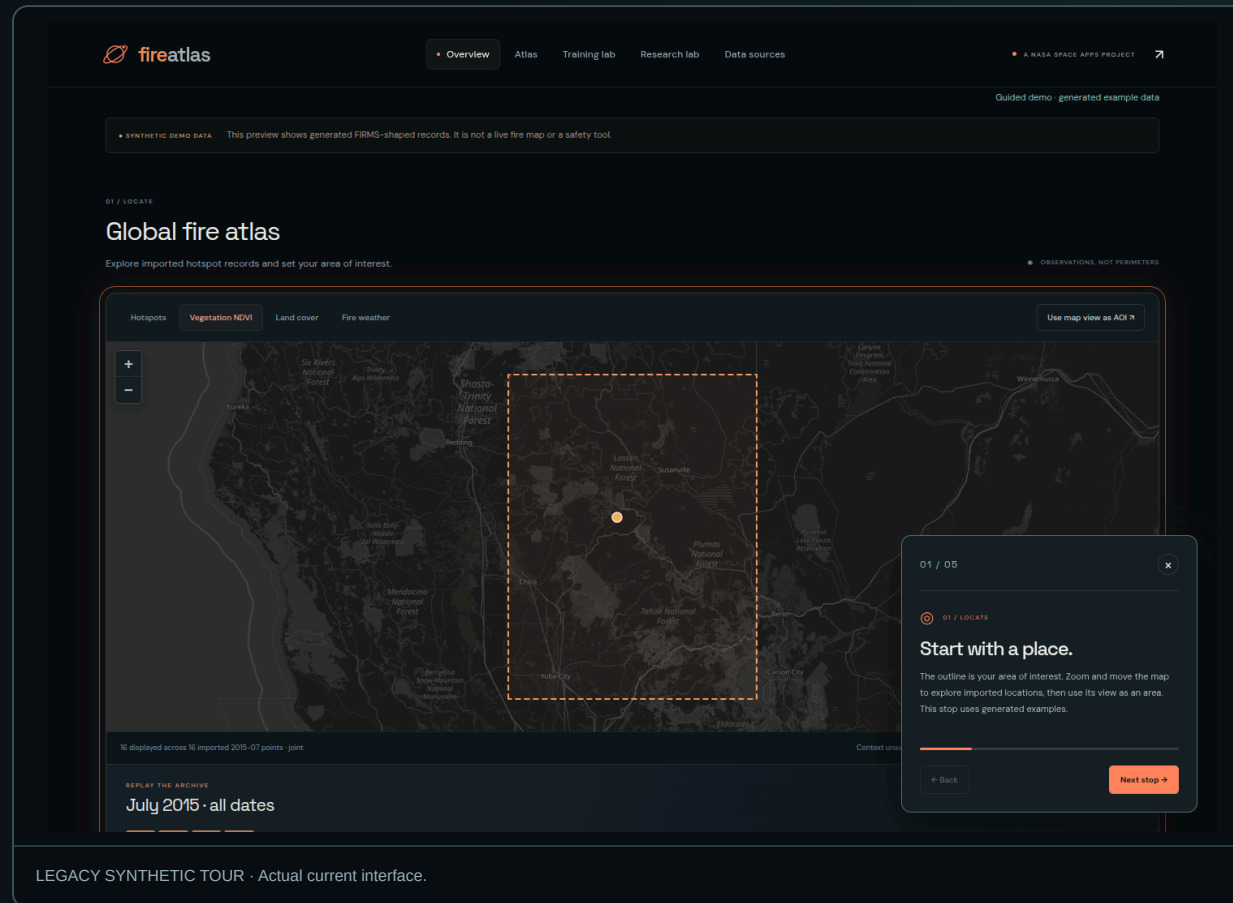
READY TO PRESENT WHEN

A saved view restores the same selection and the downloaded ZIP passes the independent verifier.

Saved views are local to one browser. Links require access to the same server. Hashes check integrity, not authenticity.

PRIORITY 14 / LEGACY SYNTHETIC TOUR

The five-stop guided story



LEGACY SYNTHETIC TOUR · Actual current interface.

EXACT PAGE / OPEN THIS VIEW<http://127.0.0.1:8000/#calendar-section>**Interaction:** Click Start guided tour → follow all five stops → open Training Lab.

Why it stands out

The walkthrough teaches the product by operating the real controls used elsewhere on the site.

BEAUTIFY NEXT

- Place each tour card close to its target without covering it.
- Keep progress, next and back controls consistent.
- Trim each stop to one idea and one visible change.

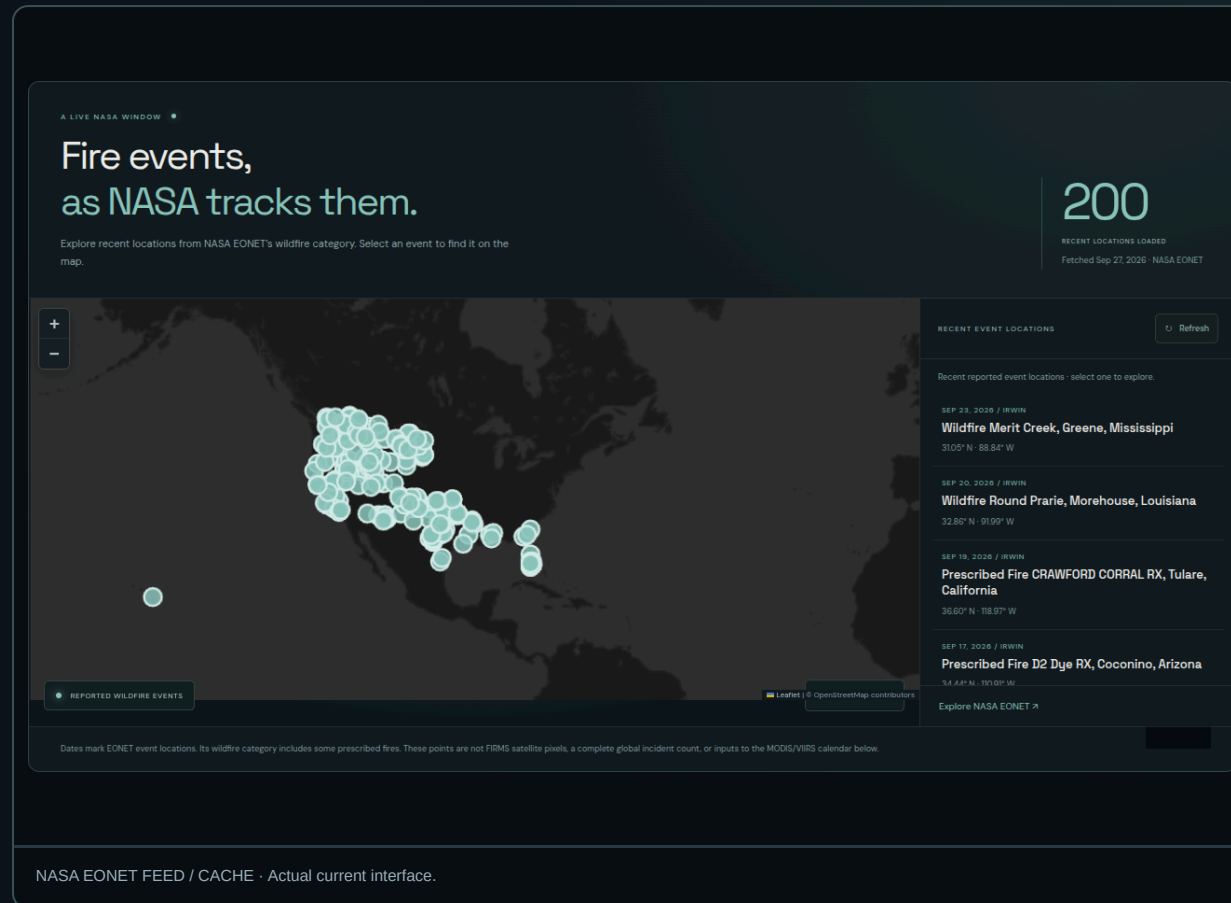
READY TO PRESENT WHEN

A first-time user completes the five stops without help and can repeat the sensor comparison independently.

The tour deliberately uses a small July 2015 example: four MODIS pixels, twelve VIIRS pixels and four joint cell-days.

PRIORITY 15 / NASA EONET FEED / CACHE

A separate window onto reported events



Why it stands out

Named reported events provide a recognizable entry point alongside the satellite observation tools.

BEAUTIFY NEXT

- Strengthen selection feedback between the event list and map.
- Make update time and cached-versus-connected status prominent.
- Keep event cards short with a consistent title, date and source.

READY TO PRESENT WHEN

Selecting a report visibly locates it and the presenter can explain how reports differ from satellite pixels.

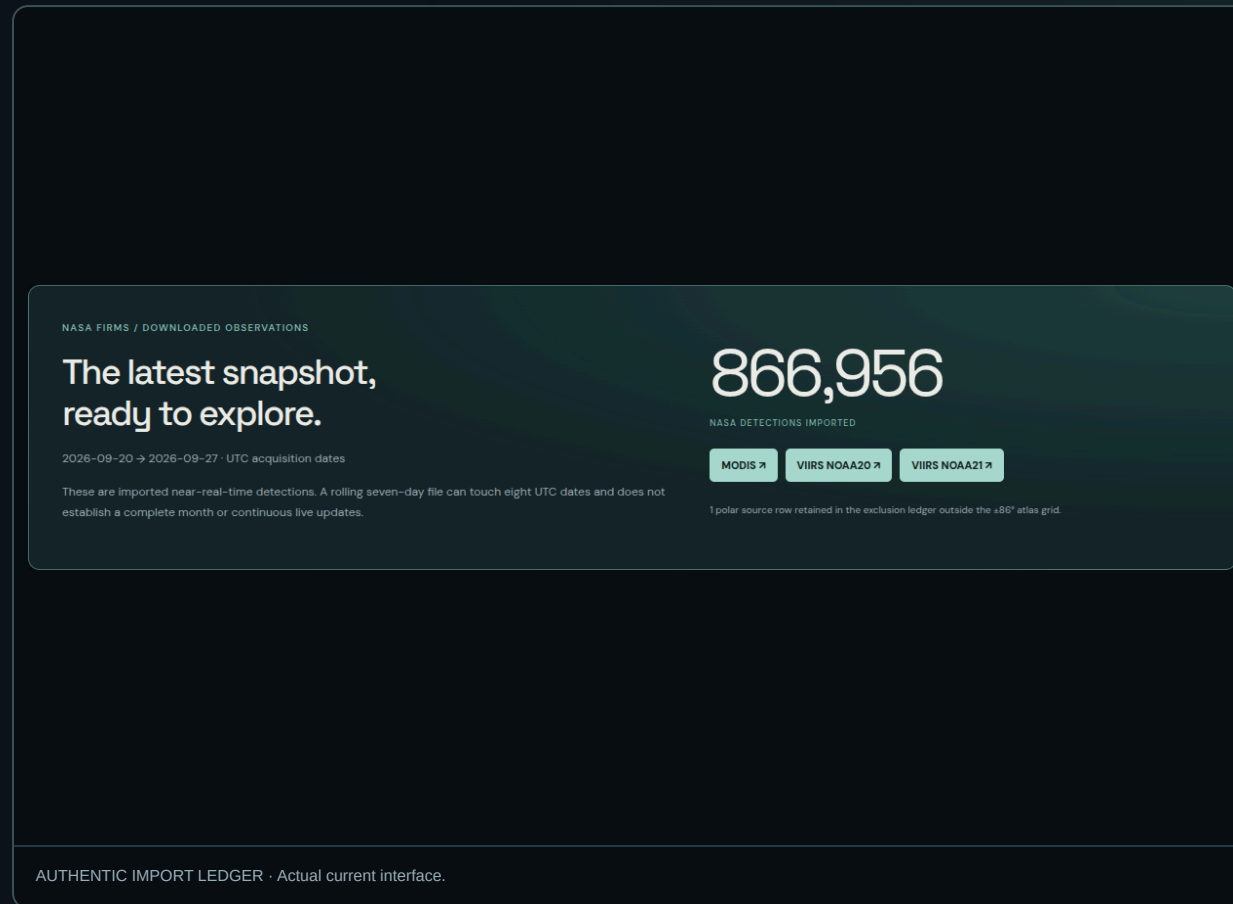
EONET reports remain separate from FIRMS counts. The feed may contain prescribed burns and is not a complete global incident register.

EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/#live-events>

Interaction: Choose a reported event → locate it on the map → inspect the event source and update time.

Data provenance that a judge can inspect



EXACT PAGE / OPEN THIS VIEW

<http://127.0.0.1:8000/data.html>

Interaction: Inspect the recent CSV snapshot → follow a satellite link → contrast with the separate synthetic showcase.

Why it stands out

The project shows where its data came from and makes the demonstration's generated inputs explicit.

BEAUTIFY NEXT

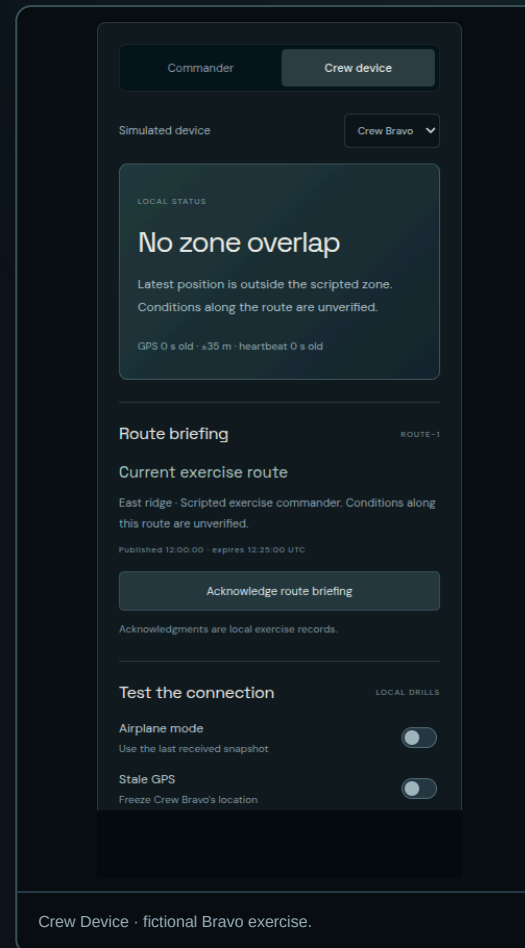
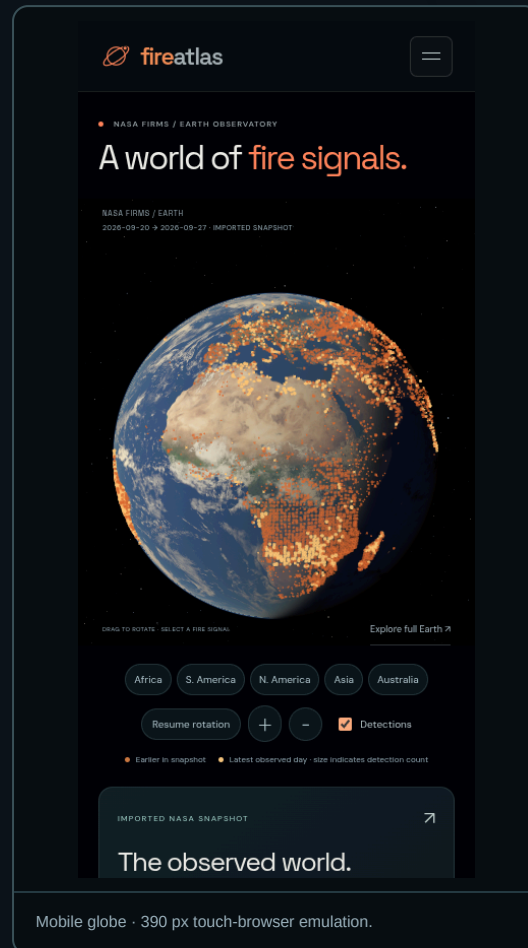
- Use clear Authentic / Synthetic / Cached labels consistently across pages.
- Lead with useful source facts; keep hashes and detailed logs expandable.
- Turn imported-versus-missing coverage into a readable status summary.

READY TO PRESENT WHEN

A judge can identify which visuals use authentic observations and which use demonstration fixtures.

916,376 authentic records total: 866,956 NASA + 49,420 NOAA. One polar NASA row is retained in the exclusion ledger.

What is the mobile app today?



Now: an installable mobile web app.

The responsive website already provided the globe, atlas, research tools and Crew Device exercise. This update adds a web-app manifest, icons, standalone launch, installation instructions and an explicit offline reconnect screen.

FUNCTIONAL BROWSER CHECKS

Mobile crew selection, replay, prepared-tab offline reload, saved time/crew restoration and offline exercise export passed. The narrower Training Lab worker continues to own its offline behavior.

NOT A NATIVE APP RELEASE

No Android APK or iOS native project exists. No real device GPS, push alerts, background tracking or shared commander/crew sessions are implemented.

Phone presentation: use a reachable HTTPS deployment, install it from the phone's browser, and rehearse on that physical phone.

Browser emulation and installability validation are not a physical-device installation test. The globe and analysis tools still require the application server.

<http://127.0.0.1:8000/install.html>

Get the phone ready before rehearsal.

1. Reach the same dataset

Deploy the Python application and its database at a trusted HTTPS address reachable by the phone. A GitHub repository is not the running backend.

Copy the populated local database or import the original NASA CSVs on that host. A fresh checkout includes the NOAA sample and synthetic seeds; it does not include the 866,956 recent global NASA records.

Do not open localhost:8000 on the phone. That points to the phone itself. For an online LAN-only preview, run the server with `- host 0.0.0.0` on a trusted network and use the laptop's LAN address. Plain HTTP LAN access does not provide secure PWA installation or offline service workers.

2. INSTALL FROM THE BROWSER

Android Chrome: menu → Install app / Add to Home screen.

iPhone Safari: Share → Add to Home Screen → Open as Web App if offered.

MDN: installation requirements · Apple: home-screen instructions

3. Prepare the offline demonstration

1. Open Training Lab while connected.
2. Wait for **Offline page ready** and **Progress saved in this tab**.
3. Advance to 12:05, choose Crew Bravo and keep this tab open.
4. Disconnect, reload the same tab and show the saved clock plus disconnected status.
5. Export the exercise record offline, reconnect and resume.

4. CHECK THE ACTUAL PHONE

Confirm readable controls, touch rotation, satellite filtering, crew selection, browser downloads, installation and same-tab offline recovery.

No public HTTPS deployment or physical-phone test was completed during this documentation task. Those are the remaining setup gates for presenting from a separate phone.

A five-minute presentation path.

1. **0:00–0:45 · Earth.** Rotate the globe, point out the imported snapshot date, select one cluster and inspect its evidence.
2. **0:45–1:45 · Calendar + sensors.** Open the September 2026 synthetic showcase. Show the seasonal peak, three-year baseline and MODIS / VIIRS / Joint comparison.
3. **1:45–2:15 · Traceability.** Open a detected day, expand a record, then show Download study. Explain how exported inputs reproduce the counts.
4. **2:15–3:15 · Research.** Show the overlap chart, three candidate groups and the synthetic exposure mask. Clear the mask to demonstrate why unknown coverage matters.
5. **3:15–4:30 · Mobile crew.** Open the fictional Crew Device view, advance the replay and demonstrate the disconnection drill or prepared-tab offline reload.
6. **4:30–5:00 · Close with evidence.** Return to Data Sources. State which inputs are authentic, which are synthetic, and the next visual improvement.

The strongest story is: **see a signal** → **compare fairly** → **inspect the evidence** → **rehearse a decision.**

Keep SAR, calibrated sensitivity, operational spread forecasts and real crew tracking off the feature demo until implemented and validated. They are future requirements, not available capabilities.

What was checked—and what remains.

Verified in this workspace

- Current globe: 866,956 authentic NASA detections, 7,307 one-degree display clusters.
- Authentic database: 916,376 observations including the NOAA historical slice.
- Separate full-year synthetic showcase: 11,322 records across 96 source-months.
- All 56 automated tests and the application build passed after the mobile-app addition.
- Fresh desktop and mobile screenshots were captured for this guide.
- Mobile browser checks covered offline launch, training-worker ownership, same-tab reload, saved crew/clock and offline export.

The matching code verification log and screenshot manifest accompany this guide. These checks verify application behavior, not scientific validity or real-world crew safety.

Still requires a real input or deployment

- A trusted HTTPS host reachable from the presentation phone, with the intended database loaded.
- A rehearsal on the actual Android/iPhone and browser.
- Native packaging if an APK or iOS build is specifically required.
- Real pass/cloud masks, measured weather, validated SAR analysis and calibrated spread models for scientific claims.
- Real cross-device sessions, permissions, GPS and background delivery for operational crew use.

USEFUL PROJECT REFERENCES

PRD.md · docs/DATA_READINESS.md · docs/GLOBE_OBSERVATIONS.md · docs/RESEARCH_METHODS.md

Open Data Sources · Open mobile setup

All page links use the current local server. Replace `http://127.0.0.1:8000` with the presentation host when sharing with another device. Screenshots and captions identify synthetic, authentic and fictional content.